
MAE Preliminary Examination

Mathematics Section

Wednesday, November 13, 2024, 1:00pm-3:30pm

Your Name

THREE PROBLEMS WILL BE GRADED	
Select the 3 problems you've worked, to be graded:	points
Problem:	/10
Problem:	/10
Problem:	/10
Total	/30

Please give your answers/work in the space provided
Explain your work/steps clearly

Linear Algebra: Problem 1 [10 points]: This problem has 5 subproblems for 2 points each.

1. Suppose three square matrices of the same size, A, B, T relate as follows

$$AT = TB,$$

and that $\det(T) \neq 0$. **What do A and B have in common?**

2. Suppose A, B are symmetric matrices of the same size, and that $A = TBT'$, where T' denotes transpose, and that T is invertible. **What do A and B have in common?**
3. **Give conditions** on a matrix A and a vector b so that the equation $Ax = b$ **has a unique solution**.
4. If c_k ($k \in \{1, \dots, 5\}$) are scalars so that $v = \sum_{k=1}^5 c_k w_k$, and

$$v = \begin{bmatrix} 6 \\ 0 \\ 4 \\ 2 \\ 8 \end{bmatrix}, w_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, w_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, w_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, w_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}, w_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix},$$

find c_4 .

5. Consider two positive semi-definite matrices A, B of the same size $n \times n$, and denote by a_{ij}, b_{ij} , for $i, j \in \{1, \dots, n\}$, their respective elements; that is, $A = [a_{ij}]_{i,j=1}^n$ and $B = [b_{ij}]_{i,j=1}^n$. **Show that the matrix $C = [a_{ij}b_{ij}]_{i,j=1}^n$ is also positive semi-definite.**

Hint: Any $n \times n$ positive semi-definite matrix M can be written as $M = \sum_{k=1}^n \mu_k m_k m_k^T$, where m_k is the (column) eigenvectors corresponding to the eigenvalue μ_k . Moreover, it suffices to show that the statement is true when B has rank 1.

Workspace for Problem 1: Explain your reasoning/work here.

Solution:

- They are similar, since $A = TBT^{-1}$, so they share the **same eigenvalues and Jordan structure**.
- They are congruent, so they share the **same number of positive, negative and zero eigenvalues**. That is, they have the same signature.
- $\text{rank}([A, b]) = \text{rank}(A)$ ensures that the equation **has a solution**, while $\text{rank}(A) =$ to the number of columns of A ensures that there are no null vectors $Ax = 0$ and thus, **the solution is unique**.
- $c_4 = 3$. E.g., you can observe that $c_4 + c_5 = 8$ while $-c_4 + c_5 = 2$. Alternatively, you can observe that the w_k 's are orthogonal. Hence, $w'_4 v = c_4 w'_4 w_4$, since $w'_4 w_1 = 0$ etc., giving the same result.
- Assume that B has rank 1. Then $B = \beta b b^T$, with $\beta > 0$ and $b = [b_\ell]_{\ell=1}^n$ being a column vector. Then

$$C = \beta [a_{ij} b_i b_j] = \beta \times \text{diagonal}(b_1, \dots, b_n) \times A \times \text{diagonal}(b_1, \dots, b_n),$$

which is congruent to A and hence non-negative definite.

If B has rank larger than 1, then repeat for every dyad in the dyadic decomposition of B .

Differential Equations: Problem 2 [10 points]:

For the stationary nonlinear system

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2^2 \\ x_2 - x_1x_2 \end{bmatrix}$$

Do the following:

1. Find all equilibrium states, $x_{eq,i}$.
2. Determine the linearized dynamics at each equilibrium.
3. Determine the eigenvalues of each linearized system, and classify the equilibria.
4. Sketch the phase portrait.

Solution:

Equilibria satisfy

$$x_1 - x_2^2 = 0, \quad x_2 - x_1x_2 = x_2 - x_2^3 = 0.$$

Thus

$$x_2 \in$$

We consider the two cases:

- If $x_2 = 0$, then $x_1 = x_2^2 = 0$, giving the equilibrium

$$(0, 0).$$

- If $x_1 = 1$, then $1 = x_2^2$, so $x_2 = \pm 1$, giving

$$(1, 1), \quad (1, -1).$$

Therefore the equilibrium points are

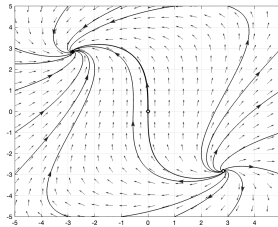
$$(0, 0), \quad (1, 1), \quad (1, -1).$$

1. Two equilibria: $x_1 = x_2 = +2\sqrt{2}$, and $x_1 = x_2 = -2\sqrt{2}$.
2. The linearized dynamics correspond to the matrix

$$A = \begin{bmatrix} 2x_1 & -2x_2 \\ x_2 & x_1 - 2x_2 \end{bmatrix} = \begin{bmatrix} 2a & 2a \\ -a & 3a \end{bmatrix}$$

for $a = \pm\sqrt{8}$, in the two cases.

3. The eigenvalues are $7.0711 \pm 3.7417i$ in the first instance, when $x_1 > 0$, and $-7.0711 \pm 3.7417i$ for the second equilibrium where $x_1 < 0$. Thus, the first is an unstable focus, and the second a stable one.



4. The phase portrait is given above.

Workspace for Problem 2: Explain your reasoning/work here.

Solution:

Partial Differential Equations: Problem 3 [10 points]:

Use the Fourier transform to solve the heat equation

$$u_t - \alpha u_{xx} = 0$$

for the semi-infinite domain $0 \leq x < \infty$ and $t \geq 0$ with $u(t, 0) = 0$ and $u(0, x) = f(x)$, given that

$$f(x) = \begin{cases} 1, & \text{for } 1 < x < 2 \\ 0, & \text{otherwise.} \end{cases}$$

Workspace for Problem 3: Explain your reasoning/work here.

Solution:

(1)

Solution to Problem 3 using Fourier transform

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} ; t \geq 0, x \geq 0$$

$$u(t, 0) = 0 ; u(0, x) = f(x) = H(x-1) - H(x-2)$$

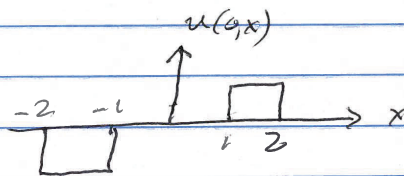
H is the heaviside function.

Fourier transforms are natural for infinite domains. We must create a contrived problem with infinite x -domain in order to solve our actual semi-infinite domain problem.

In order to maintain $u(t, 0) = 0$, we must have an anti-symmetric formulation for the infinite domain.

Namely, we set $u(0, x) = -f(-x)$ for $x < 0$ but maintain $u(0, x) = f(x)$ for $x \geq 0$. Now, we have the infinite domain problem

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} ; u(0, x) = H(x-1) - H(x-2) - H(1-x) + H(2-x)$$
$$t \geq 0, -\infty < x < \infty$$



(2)

Transforms

$$\hat{u}(t, \xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\xi x} u(t, x) dx$$

$$\frac{\partial \hat{u}}{\partial t} = \frac{\partial \hat{u}}{\partial t} ; \frac{\partial \hat{u}}{\partial x} = i\xi \hat{u} ; \frac{\partial^2 \hat{u}}{\partial x^2} = -\xi^2 \hat{u}$$

$$\frac{\partial \hat{u}}{\partial t} = -\alpha \xi^2 \hat{u} ; \hat{u}(0, \xi) = \hat{f}(\xi)$$

Solution to ODE $\hat{u} = \hat{f}(\xi) e^{-\alpha \xi^2 t}$

The inverse transform of $e^{-\alpha \xi^2 t}$ is

$$g(t, x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\xi x} e^{-\alpha \xi^2 t} d\xi$$

$$= \frac{1}{\sqrt{4\pi\alpha t}} e^{-x^2/4\alpha t} \quad \text{by integration}$$

Using the convolution theorem

$$u(t, x) = \frac{1}{\sqrt{4\pi\alpha t}} \int_{-\infty}^{\infty} e^{-(x-x')^2/4\alpha t} f(x') dx'$$

Now, using the function for $f(x')$, we have

$$u(t, x) = \frac{1}{2\sqrt{\pi\alpha t}} \left[\int_1^2 e^{-(x-x')^2/4\alpha t} dx' - \int_{-2}^{-1} e^{-(x-x')^2/4\alpha t} dx' \right]$$

Using the error function definition, we obtain

$$\operatorname{erf}(z) \equiv \frac{2}{\sqrt{\pi}} \int_0^z e^{-(z')^2} dz'$$

$$u(t, x) = \frac{1}{2} \left[\operatorname{erf}\left(\frac{2-x}{\sqrt{4\alpha t}}\right) - \operatorname{erf}\left(\frac{1-x}{\sqrt{4\alpha t}}\right) + \operatorname{erf}\left(\frac{1+x}{\sqrt{4\alpha t}}\right) - \operatorname{erf}\left(\frac{2+x}{\sqrt{4\alpha t}}\right) \right]$$

Workspace for Problem 3: Explain your reasoning/work here.

Partial Differential Equations: Problem 4 [10 points]:

Consider the wave equation $v_{tt} - \nabla^2 v = 0$ in the 2D volume $0 \leq x \leq 1$ and $0 \leq y \leq 1$ with the boundary conditions

$$v_x(t, 0, y) = 0, \quad v_y(t, x, 0) = 0, \quad v(t, 1, y) = 0, \quad v(t, x, 1) = 0$$

and the initial conditions

$$v(0, x, y) = f(x, y), \quad v_t(0, x, y) = g(x, y).$$

- (a) Relate $v(t, x, y)$ to $u(x, y)$ that satisfy the 2D Helmholtz equation $\nabla^2 u + \lambda u = 0$ over the domain $0 \leq x \leq 1$ and $0 \leq y \leq 1$, where λ is an eigenvalue which must be determined. The boundary conditions are:

$$u_x(t, 0, y) = 0, \quad u_y(t, x, 0) = 0, \quad u(t, 1, y) = 0, \quad \text{and} \quad u(t, x, 1) = 0$$

- (b) Determine the eigenvalues λ_n for $n = 1, 2$, etc.
(c) Determine the corresponding eigenfunctions, u_n for $n = 1, 2$, etc.
(d) Determine the coefficients that enter the relation between $v(t, x, y)$ and $u_n(x, y)$.

Workspace for Problem 4: Explain your reasoning/work here.

Solution:

Assume that the solution u the eigenvalue problem $\nabla^2 u + \lambda u = 0$ in the separable form $u(x, y) = X(x)Y(y)$. We then have

$$X''Y + XY'' + \lambda XY = 0,$$

which implies

$$\frac{X''}{X} + \frac{Y''}{Y} = -\lambda \quad \text{and} \quad \frac{X''}{X} = -\frac{Y''}{Y} - \lambda = -\omega^2, \quad (1)$$

for some constant ω .

The solution of the X -equation

$$X'' + \omega^2 X = 0$$

is given by

$$X(x) = A \cos(\omega x) + B \sin(\omega x).$$

Since $X'(0) = 0$, we have $B = 0$; and $X(1) = 0$ implies $\omega_n = \frac{2n-1}{2}\pi$ for $n = 1, 2, \dots$

Similarly, the solution of the Y -equation

$$Y'' + (\lambda - \omega^2)Y = 0$$

is given by

$$Y(y) = A \cos(\sqrt{\lambda - \omega^2} y) + B \sin(\sqrt{\lambda - \omega^2} y).$$

Since $Y'(0) = 0$, we have $B = 0$; and $Y(1) = 0$ implies

$$\lambda_{mn} = \left(\frac{2m-1}{2}\pi \right)^2 + \left(\frac{2n-1}{2}\pi \right)^2.$$

As such, the eigenfunctions u_{mn} are written as

$$u_{mn}(x, y) = \cos(\omega_n x) \cos(\sqrt{\lambda_{mn} - \omega_n^2} y) = \cos\left(\frac{2n-1}{2}\pi x\right) \cos\left(\frac{2m-1}{2}\pi y\right)$$

Now, consider the solution v of the wave equation and assume it in the separable form

$$v(x, y, t) = X(x)Y(y)T(t).$$

Substituting this form into the wave equation, we obtain

$$T''XY - TX''Y - TXY'' = 0,$$

implying

$$\frac{T''}{T} = \frac{X''}{X} + \frac{Y''}{Y} = -\lambda \quad \text{and} \quad \frac{X''}{X} = -\frac{Y''}{Y} - \lambda = -\omega^2 \quad (2)$$

for some constants λ and ω , to be determined. This equation is identical to Eq. (1), governing the eigenfunctions in the wave equation problem. Moreover, since the two problems have the same boundary conditions, the mode shapes will be identical. That is, the spatial part of the solution v to the wave equation will be composed of linear combinations of the eigenfunctions u_{mn} of the eigenvalue problem:

$$v(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} T_{mn}(t) u_{mn}(x, y),$$

where the temporal part T_{mn} satisfies

$$T_{mn}'' + \lambda_{mn} T_{mn} = 0$$

whose solution is given by

$$T_{mn}(t) = A_{mn} \cos(\sqrt{\lambda_{mn}} t) + B_{mn} \sin(\sqrt{\lambda_{mn}} t).$$

As such, the total solution v of the wave equation is written as

$$v(t, x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left[A_{mn} \cos(\sqrt{\lambda_{mn}} t) + B_{mn} \sin(\sqrt{\lambda_{mn}} t) \right] \cos\left(\frac{2n-1}{2}\pi x\right) \cos\left(\frac{2m-1}{2}\pi y\right)$$

The initial conditions then give

$$v(0, x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \cos\left(\frac{2n-1}{2}\pi x\right) \cos\left(\frac{2m-1}{2}\pi y\right) = f(x, y),$$

which implies that A_{mn} are the coefficients of the Fourier cosine series representation of the function $f(x, y)$. Hence, they are given by

$$A_{mn} = 4 \int_0^1 \int_0^1 f(x, y) \cos\left(\frac{2n-1}{2}\pi x\right) \cos\left(\frac{2m-1}{2}\pi y\right) dx dy.$$

Similarly, the initial condition on velocity dictates

$$v_t(0, x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sqrt{\lambda_{mn}} \cos\left(\frac{2n-1}{2}\pi x\right) \cos\left(\frac{2m-1}{2}\pi y\right) = g(x, y),$$

which implies that $B_{mn} \sqrt{\lambda_{mn}}$ are the coefficients of the Fourier cosine series representation of the function $g(x, y)$. Hence, they are given by

$$B_{mn} = \frac{4}{\sqrt{\lambda_{mn}}} \int_0^1 \int_0^1 g(x, y) \cos\left(\frac{2n-1}{2}\pi x\right) \cos\left(\frac{2m-1}{2}\pi y\right) dx dy.$$

Workspace for Problem 4: Explain your reasoning/work here.